FTXL 3190 Free Topology Transceiver
Models 14260R-800, 14240R, and 14250R-300



- Highest performance ANSI/CEA-709.1-B and EN14908.1 solution
- 4096 network variables with LonMark® standard changeable-type support
- Dynamic network variable support
- Supports up to 4095 address table entries
- Supports up to 200 concurrent send and receive transactions
- Supports up to 8190 Alias table entries
- Leverages scaleable Nios® II embedded processor technology of the Altera® Cyclone® II/III FPGA device families
- Shares a common platform and API with ShortStack® applications
- Supports polarity-insensitive free topology star, daisy chain, bus, loop, or mixed topology wiring
- 78 kilobits-per-second bit rate for distances up to 500 meters in free topology or 2700 meters in bus topology with double terminations
- Unique 48-bit Neuron® ID in every device for network installation and management
- Compact external transformer with patent-pending architecture providing exceptional immunity from magnetic interference and high frequency common mode noise
- Compatible with TP/FT-10 channels using devices based on FT 3120/3150 Smart Transceivers, FTT-10 and/or FTT-10A Free Topology Transceivers, and, with suitable DC blocking capacitors, LPT-10 Link Power Transceivers
- Communications parameters preprogrammed for the TP/FT-10 channel at 10MHz
- 5V operation with low power consumption
- -40 to +85°C operating temperature range[1, 2]

Description

The FTXL 3190 Free Topology Transceiver is a high-performance free topology transceiver and media access controller for developing high-performance devices based on the LonTalk Platform. The LonTalk® Platform for the FTXL transceiver combines development tools, an application programming interface (API), and a high-performance ANSI/CEA-709.1-B (EN14908.1) protocol stack with the FTXL transceiver providing unparalleled performance coupled with ease-of-development, reliability, and stability. The LonTalk Platform for FTXL transceivers is ideally suited for high-value controllers required for system and area level applications. It removes restrictions associated with address table size and transaction control and address table limitations that existed with earlier solutions. The LonTalk Platform is also available for the ShortStack® Micro Server—providing a scalable solution from any ShortStack host processor to an FTXL solution with an Altera host processor.

For maximum performance, reliability, and flexibility, Echelon has ported its widely used LonTalk protocol stack to the Nios II embedded processor supported by the Altera FPGA devices. Echelon's FTXL LonTalk protocol stack is a complete implementation of the ANSI/CEA-709.1-B (EN14908.1) Control Networking Protocol, including support for the ANSI/CEA-709.1-B (EN14908.1) Enhanced Command Set providing higher performance for controller applications. The rich mix of pin and logic element configurations offered by the Cyclone II/III family of FPGA devices combined with the high-performance FTXL LonTalk protocol stack provides developers with the options to exactly fit their application requirements for controller devices connected directly to a TP/FT-10 LonWorks® control network segment. Used in conjunction with the FTXL transceiver, the Nios II embedded processor on the Cyclone II/III FPGA provides a powerful and scaleable platform for system controller applications in LonWorks networks.

The foundation code base for the FTXL LonTalk protocol stack has already been widely deployed in Echelon's *i*.LON® and LNS® product platforms for proven reliability and performance.

The LonTalk Platform is a compatible family of development tools, an API, firmware, and chips that can be used to implement a wide variety of LonWorks devices, from simple sensors and actuators to complex system- and area-controllers. The LonTalk Platform for the FTXL transceiver provides an Altera FPGA-hosted solution with unparalleled performance for applications requiring up to 4096 network variables. The LonTalk Platform is also available for the ShortStack Micro Server—providing a solution that can be used with any host processor that requires up to 254 network variables. Both solutions share compatible application programming interfaces (APIs) and interface builder tools, lowering the learning time for developers using both frameworks, and simplifying the use of common code for both FTXL transceivers and ShortStack Micro Servers.

Notes:

[1] EEPROM programming must be limited to -25 to 85°C for a 10-year data retention over the -40 to 85°C operating temperature range.

[2] Maximum junction temperature should not exceed 105ºC. Tjunction can be calculated as follows: Tjunction = TAmbient + V•I•qJA where qJA for 32-pin SOIC = 51°C/W, qJA for 44-pin TQFP = 43°C/W, and qJA for 64-pin TQFP = 44°C/W.

The FTXL 3190 transceiver is fully compatible with the LONMARK TP/FT-10 channel and can communicate with devices implemented with Echelon's FT 3120 and 3150 Smart Transceivers, FTT-10A Free Topology Transceiver, and when used with suitable DC blocking capacitors, the FTXL transceiver is also fully compatible with the LPT-10 and LPT-11 Link Power Transceivers.

The FTXL 3190 transceiver is offered in a compact 44-lead TQFP package. It supports input clock rates of 5, 10, 20, and 40 MHz (20 MHz recommended). A communication transformer must be used with the FTXL 3190 transceiver. The FT-X1 Communication Transformer is a through-hole component while FT-X2 is a surface- mount component. Either transformer can be used with the FTXL 3190 transceiver. The FT-X1 and FT-X2 transformers have similar noise immunity and performance characteristics.

The FTXL 3190 transceiver and communication transformers (Models 14260R-800, 14240R, and 14250R-300) are compliant with the European Directive 2002/95/EC on the restriction of the use of certain hazardous substances (RoHS) in electrical and electronic equipment.

Solution Architecture

A device implemented with an FTXL transceiver includes the following components (see Figure 1):

- Application and FTXL Nios II library for a 32-bit Nios II embedded processor implemented on an Altera FPGA device
- FTXL 3190 transceiver
- FT-X1 or FT-X2 Communication Transformer

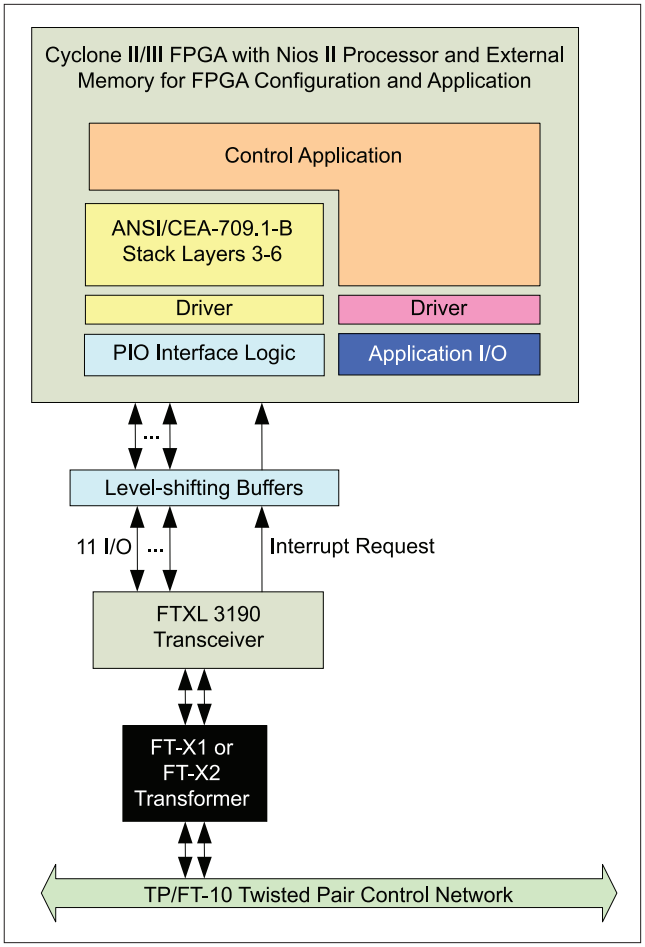


Figure 1: FTXL Solution Architecture

The interface between the Nios II host processor and FTXL 3190 transceiver uses a high-speed 8-bit parallel interface that includes support of an uplink interrupt as packets from the network are received and moved up the LonTalk protocol stack running on the Nios II host. The FTXL 3190 transceiver implements layers 1 and 2 of the ANSI/CEA-709.1-B (EN14908.1) protocol to free the Nios II processor from media access processing. The default buffer configuration on the FTXL transceiver provides support for the maximum ANSI/CEA-709.1-B (EN14908.1) packet size of 255 bytes. The device driver uses the Altera Hardware Abstraction Layer (HAL) to implement a portable interface that may be effortlessly incorporated in your design, even if your final FPGA hardware solution does not match that of the development board.

Altera Nios II FPGA Benefits

The Altera family of FPGA devices provides a rich mix of pin and logic element configurations, and the Nios II embedded processor comes in three performance-levels (economy, standard and fast) to choose from, allowing trade-offs of logic element count vs. performance. The 32-bit Nios II embedded processor may also be configured to support different memory and cache configurations to provide the performance necessary for the application. Together, Altera FPGA and Nios II embedded processor technology allow system designers to customize the peripherals and hardware resources to exactly match the application needs. Altera C to hardware compilation technology can dramatically improve system performance by implementing algorithms normally handled in software as custom instructions in hardware. Math intensive applications can benefit with the addition of hardware multipliers to the Nios II embedded processor. Altera FPGA and Nios II embedded processor technology allow system designers to customize the peripherals and hardware resources to exactly match the application needs. Altera System-on-a-Programmable-Chip (SOPC) technology allows for rapid and flexible hardware design.

Development Platform

The FTXL Developer's Kit is required to develop applications for the FTXL transceiver. The developer's kit is available as a free download from www.echelon.com and includes the following:

- FTXL Nios II library supporting layers 3-6 of the ANSI/CEA-709.1-B (EN14908.1) protocol, and interfacing with the FTXL transceiver
- Nios II HAL-compliant drivers targeting custom hardware peripherals provided with the reference design
- A utility, called the *LonTalk Interface Builder*, creates C declarations for a specified LONWORKS interface, and also creates LonTalk protocol stack structures and data tables

- ▼ Example FTXL applications built using the Altera Nios II Embedded Development Suite (EDS)
- ▼ Schematics providing reference implementations for the level-shifting interface required to connect the FTXL transceiver to the Cyclone II FPGA
- ▼ Quartus® 7.1 compatible design files to execute the examples on the DBC2C20 development board and easy migration to your own hardware design
- ▼ Source files providing an OS compatibility layer (OSCL) to allow developers to target a real-time kernel other than Micrium µC/OS-2

Design and development tools for an Altera FPGA and the Nios II soft processor are required to develop applications using the FTXL Framework. The FTXL solution has been tested with Altera's Quartus 7.2 release with SOPC Builder and the Altera Nios II Embedded Development Suite (EDS). This software is available from Altera, and is subject to Altera licenses. For more information on these tools, see www.altera.com.

Your application development is accelerated by using development hardware from Devboards (www.devboards.de). The FTXL Developer's Kit includes example applications and Cyclone II design files that target the Devboards DBC2C20 Cyclone II development board. The DBE-FT-PAR and DBE-ADAP add-on boards and the DBC2C20 development board are available from Devboards.

With the FTXL Developer's Kit, a Devboards DBC2C20 Cyclone II development board, and the FTXL Developer's Kit add-on boards, you can quickly prototype a working ANSI/CEA-709.1-B (EN14908.1) solution for your controller design with no hardware development effort.

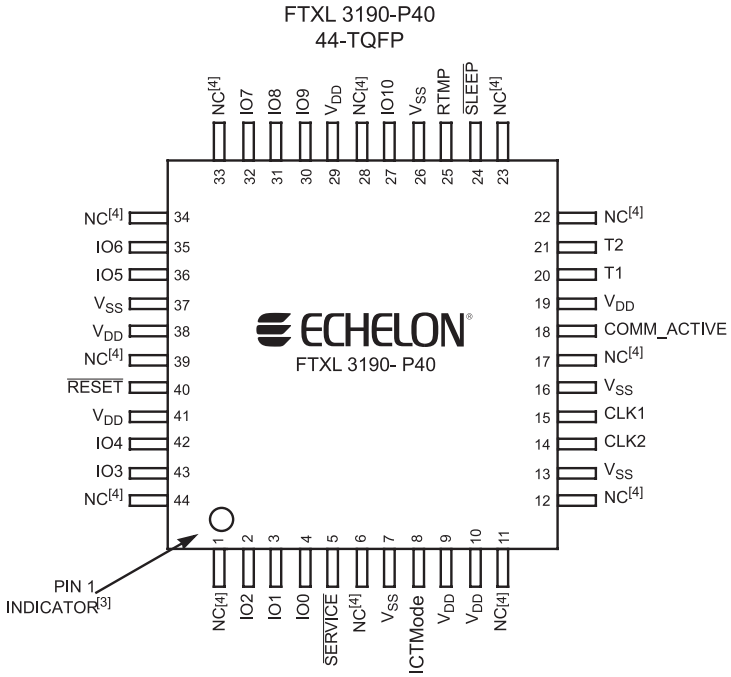
For more information on the developer's kit, see the *Model 10050-10 FTXL Developer's Kit Datasheet*.

Network Noise Protection

The communication transformer enables operation in the presence of high frequency common mode noise on unshielded twisted pair networks. Properly designed devices can meet the rigorous Level 3 requirements of EN 61000-4-6 without the need for a network isolation choke. The transformer also offers outstanding immunity from magnetic noise, eliminating the need for protective magnetic shields in most applications.

FT-X1/FT-X2 Communication Transformers must be ordered separately. See FTXL 3190 Transceiver Ordering Information for product offerings and descriptions. The FTXL 3190 Free Topology Transceiver and the FTX1/FT-X2 Communication Transformer are designed to be used as a pair and therefore must be implemented together in all designs. No transformer other than the FT-X1 or FT-X2 Communication Transformer may be used with the FTXL 3190 Free Topology Transceiver, or the transceiver warranty will be void.

FTXL 3190 Transceiver IC Pin Configurations



Notes:

[3] The small dimple at the bottom left of the marking indicates pin 1.

[4] NC (No Connect) — Should not be used. (These pins are reserved for internal testing.)

FTXL 3190 Transceiver IC Pin Descriptions

Pin Name	Type	Pin Functions	FTXL 31190-P40 TQFP-44 Pin Number
CLK1	Input	Oscillator connection or external clock input.	15
CLK2	Output	Oscillator connection. Leave open when external clock is input to CLK1. Maximum of one external HCMOS equivalent load.	14
$\overline{\text{RESET}}$	I/O (Built-in Pull-up)	Reset pin (active LOW). *Note:* The allowable external capacitance connected to the $\overline{\text{RESET}}$ pin is 100pF-1000pF.	40
$\overline{\text{SERVICE}}$	I/O (Built-in Configurable Pull-up)	Service pin (active LOW). Alternates between input and output at a 76Hz rate.	5
IO0-IO3	I/O	Large current-sink capacity (20mA). General I/O port. The output of timer/counter 1 may be routed to IO0. The output of timer/counter 2 may be routed to IO1.	4, 3, 2, 43
IO4-IO7	I/O (Built-in Configurable Pull-up)	General I/O port. The input of timer/counter 1 may be derived from one of IO4-IO7. The input to timer/counter 2 may be derived from IO4.	42, 36, 35, 32
IO8-IO10	I/O	General I/O port. May be used for serial communication under firmware control.	31, 30, 27
V_{DD}	Power	Power input (5V nom). All V_{DD} pins must be connected together externally.	9, 10, 19, 29, 38, 41
V_{SS}	Power	Power input (0V, GND). All V_{SS} pins must be connected together externally	7, 13, 16, 26, 37
ICTMode	Input	In-circuit test mode control. Driving the ICTMode high and $\overline{\text{RESET}}$ low will put the device in the In-Circuit Test mode (all pins are placed in a high impedance state).	8
T1	I/O	Analog pin to be interfaced with T1 of the external transformer. Corresponds to CP0 on Toshiba and Cypress Neuron Chips.	20
T2	I/O	Analog pin to be interfaced with T2 of the external transformer. Corresponds to CP1 on Toshiba and Cypress Neuron Chips.	21
COMM_ACTIVE	Output	May be used to monitor, transmit/receive activity. Driven high during data transmissions, driven low when receiving data and kept at high impedance otherwise.	18
$\overline{\text{SLEEP}}$	Output	$\overline{\text{SLEEP}}$. May be configured as an output to ndicate when the FT 3120 / FT 3150 is in sleep mode. Corresponds to CP3 on Toshiba and Cypress Neuron Chips.	24
RTMP	Input	Reserved for future use. Must be pulled up to 5V. Corresponds to CP4 on Toshiba and Cypress Neuron Chips.	25
NC	—	No connect. Must be left open.	1, 6, 11, 12, 17, 22, 23, 28, 33, 34, 39, 44

FT-X1 Communication Transformer Pin Configuration

6-pin through-hole transformer (top view)

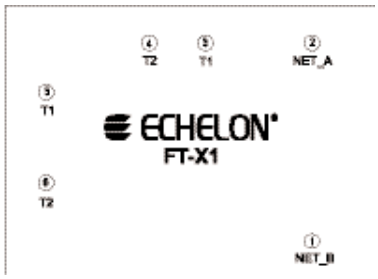


Figure 3a

FT-X2 Communication Transformer Pin Configuration

4-pin surface mount transformer (top view)



Figure 3b

FT-X1/FT-X2 Communication Transformer Pin Descriptions

Pin Name	Pin Function	Transformer Pin Number
NET_B	Network Port, polarity Insensitive	1
NET_A	Network Port, polarity Insensitive	2
T1	Internally connects to pin 5. Alternate connection to T1 pin on the FTXL 3190 IC.	3 (Not used on FT-X2)
T2	Internally connects to pin 6. Alternate connection to T2 pin on the FTXL 3190 IC.	4 (Not used on FT-X2)
T1	Connects to the ESD/transient protection circuitry and T1 pin on the FTXL 3190 IC. Internally connects to pin 3 of the FT-X1.	5
T2	Connects to the ESD/transient protection circuitry and T2 pin on the FTXL 3190 IC. Internally connects to pin 4 of the FT-X1.	6

Electrical Characteristics (VDD = 4.75-5.25V)

Parameter	Description	Min.	Max.	Unit
V_{IL}	Input Low Voltage IO0-IO10, $\overline{\text{SERVICE}}$, D0-D7, $\overline{\text{RESET}}$		 0.8	V
V_{IH}	Input High Voltage IO0-IO10, $\overline{\text{SERVICE}}$, D0-D7, $\overline{\text{RESET}}$	 2.0		V
V_{OL}	Low-Level Output Voltage I_{out} < 20µA Standard Outputs (I_{OL} = 1.4 mA)[5] High Sink (IO0-IO3), $\overline{\text{SERVICE}}$, $\overline{\text{RESET}}$ (I_{OL} = 20 mA) High Sink (IO0-IO3), $\overline{\text{SERVICE}}$, $\overline{\text{RESET}}$ (I_{OL} = 10 mA) Maximum Sink (COMM_ACTIVE) (I_{OL} = 40 mA) Maximum Sink (COMM_ACTIVE) (I_{OL} = 15 mA)		 0.1 0.4 0.8 0.4 1.0 0.4	V
V_{OH}	High-Level Output Voltage I_{out} < 20µA Standard Outputs (I_{OH} = -1.4 mA)[5] High Sink (IO0-IO3), $\overline{\text{SERVICE}}$ (I_{OH} = -1.4 mA) Maximum Sink (COMM_ACTIVE) (I_{OL} = -40 mA) Maximum Sink (COMM_ACTIVE) (I_{OL} = -15 mA)	 V_{DD} - 0.1 V_{DD} - 0.4 V_{DD} - 0.4 V_{DD} - 1.0 V_{DD} - 0.4		V
V_{hys}	Hysteresis (Excluding CLK1)	175		mV
I_{in}	Input Current (Excluding Pull-ups) (V_{SS} to V_{DD})[6]		+/- 10	µA
I_{pu}	Pull-up Source Current (V_{out} = 0 V, Output = High-Z)[6]	60	260	µA
I_{DD}	Operating Mode Supply Current [7,8] 40MHz Clock 20MHz Clock 10MHz Clock 5MHz Clock	$I_{DD(receive)}$ $I_{DD(transmit)}$ $I_{DD(receive)}$ $I_{DD(transmit)}$ $I_{DD(receive)}$ $I_{DD(transmit)}$ $I_{DD(receive)}$ $I_{DD(transmit)}$	60 75 42 57 35 50 20 35	mA mA mA mA mA mA mA mA

LVI Trip Point (VDD)

Part Number	Min.	Typ.	Max.	Unit
FTXL 3190	3.8	4.1	4.4	V

Notes:

[5] Standard outputs are IO4-IO10. ($\overline{\text{RESET}}$ is an open drain input/output. CLK2 must have 15pF load.) For FT 3150, standard outputs also include A0-A15, D0-D7, $\overline{\text{E}}$, and R/$\overline{\text{W}}$.

[6] IO4-IO7 and $\overline{\text{SERVICE}}$ have configurable pull-ups. $\overline{\text{RESET}}$ has a permanent pull-up.

[7] Supply current measurement conditions: all outputs under no-load conditions, all inputs 0.2V or (VDD - 0.2V), configurable pull-ups off and crystal oscillator clock input disabled.

[8] Maximum supply current values are at midpoint of supply voltage range.

Recommended FTXL 3190 Transceiver IC Pad Layout

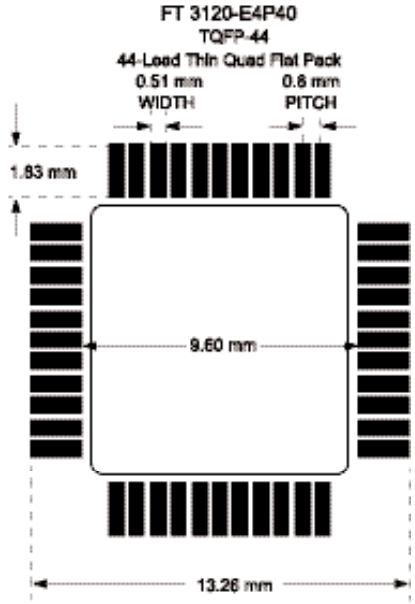


Figure 4

Recommended FT-X2 Pad Layout (4 pins)

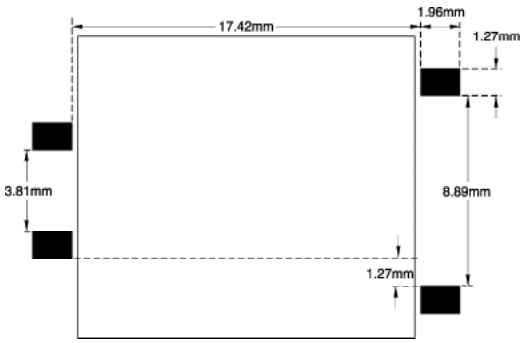


Figure 5

FTXL 3190 Transceiver IC Package Diagrams

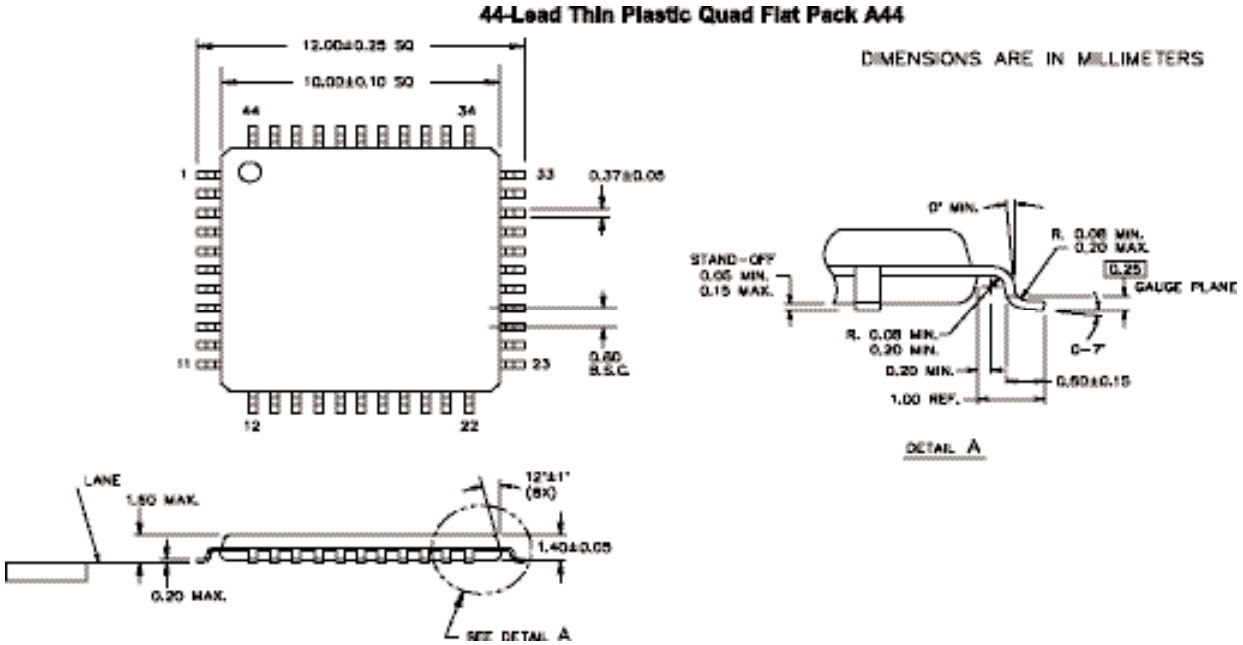


Figure 6

FT-X1 Communication Transformer Top View

(Dimensions in mm)

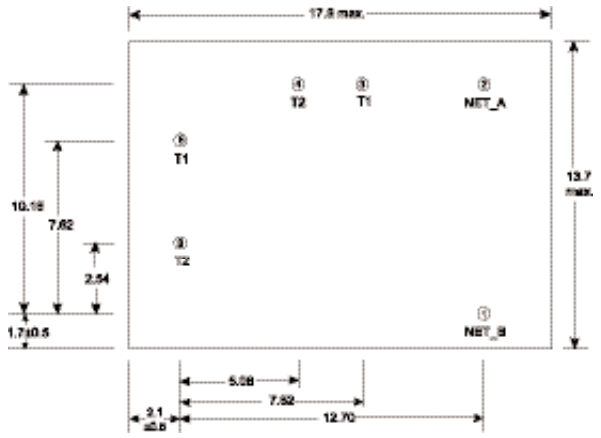


Figure 7a

FT-X1 Communication Transformer Side View

(Dimensions in mm)

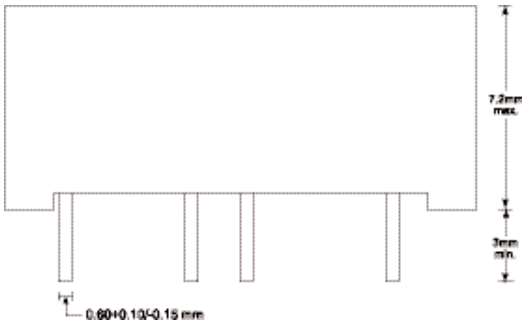


Figure 7b

FT-X2 Communication Transformer SMT Package Diagram

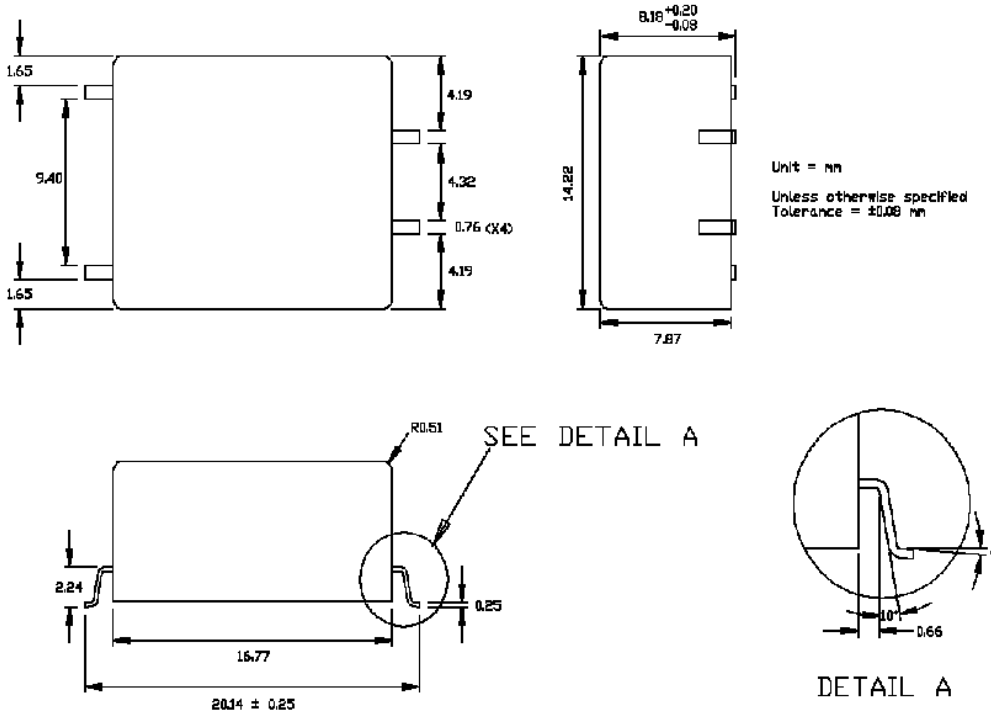


Figure 8

General Specifications

Data Communications Type	Differential Manchester coding
Network Polarity	Polarity insensitive
Isolation Between Network and	
0-60Hz, 60 seconds	1000Vrms
0-60Hz, continuous	277Vrms[9]
EMI	Designed to comply with FCC Part 15 Level B and EN55022 Level B
ESD	Designed to comply with EN 61000-4-2, Level 4
Radiated Electromagnetic Susceptibility	Designed to comply with EN 61000-4-3, Level 3
Fast Transient/Burst Immunity	Designed to comply with EN 61000-4-4, Level 4
Surge Immunity	Designed to comply with EN 61000-4-5, Level 3
Conducted RF Immunity	Designed to comply with EN 61000-4-6, Level 3
Safety Approvals (FT-X1/FT-X2 Communication Transformer)	Recognized by UL to UL Standards 60950, 2000; CSA C22.2 No. 60950, 2000; and TÜV EN 60950
Transmission Speed	78 kilobits per second
Number of Transceivers Per Segment	Up to 64
Network Wiring	24 to 16AWG twisted pair; see user's guide or *Junction Box and Wiring Guidelines* application note for qualified cable types
Network Length in Free Topology[10]	1000m (3,280 feet) maximum total wire with one repeater 500m (1,640 feet) maximum total wire with no repeaters 500m (1,640 feet) maximum device-to-device distance
Network Length in Doubly Terminated Bus Topology[10]	5400m (17,710 feet) with one repeater 2700m (8,850 feet) with no repeaters
Maximum Stub Length in Doubly-Terminated Bus Topology	3m (9.8 feet)
Network Termination	One terminator in free topology; two terminators in bus topology
Power-down Network Protection	High impedance when unpowered
Physical Layer Repeater	The FTXL 3190 transceiver cannot be used to implement a physical layer repeater. In the event that the limits on the number of transceivers or total wire distance are exceeded, FTT-10A transceivers may be used to create physical layer repeaters. See the *FTT-10A Free Topology Transceiver User's Guide* for more details.
Operating Temperature	-40 to 85°C[1]
Operating Humidity	25-90% RH @50°C, non-condensing
Non-operating Humidity	95% RH @ 50°C, non-condensing
Vibration	1.5g peak-to-peak, 8Hz-2kHz
Mechanical Shock	100g (peak)
Reflow Soldering Temperature Profile	Refer to Joint Industry Standard document *IPC/JEDEC J-STD-020C* (July 2004)
Peak Reflow Soldering Temperature	260ºC (Model 14260R-800) 260ºC (FT-X2 Model 14250R-300)

Notes:

[9] Safety agency hazardous voltage barrier requirements are not supported.

[10] Network segment length varies depending on wire type. See *Junction Box and Wiring Guidelines* application note for detailed specifications.

Ordering Information (Note: The FTXL 3190 Transceiver IC and the FT-X1/FT-X2 Communication Transformer must be ordered in the same quantities.)

The FTXL Developer's Kit is available for free download from www.echelon.com/ftxl. Contact your local Echelon representative or distributor for details on ordering the FTXL transceiver and transformer.

Product	Model Number
FTXL Developer's Kit	10050-10
FTXL 3190-P40 Transceiver Chip	14260R-800
FT-X1 Communication Transformer	14240R
FT-X2 Communication Transformer	14250R-300

Documentation

The *FTXL 3190 Transceiver DataBook* will be available soon from www.echelon.com.

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